ENGR 691-F: Assignment 2 Report

Faraz Davachi- 40305725

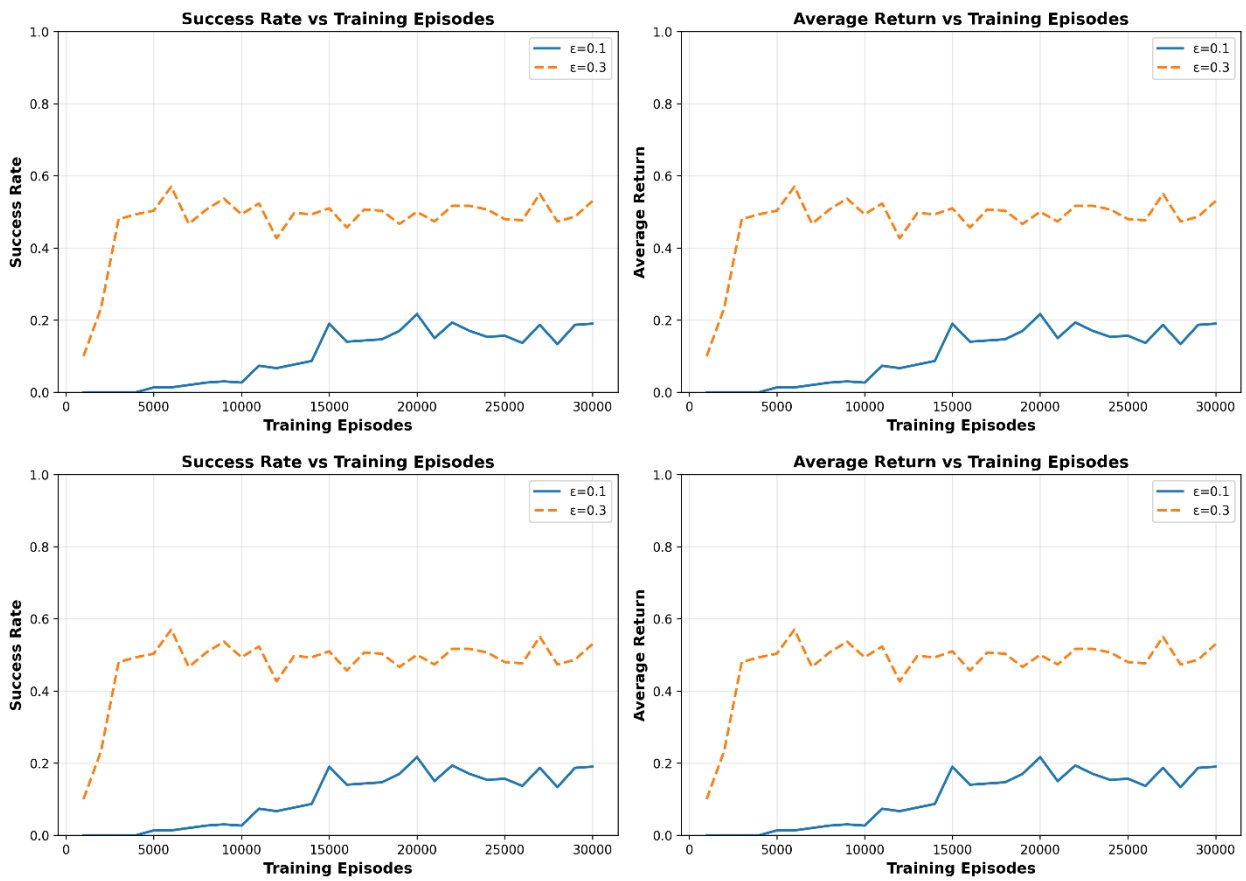
In this assignment, I implemented on-policy first-visit Monte Carlo control for the slippery FrozenLake-v1 environment in Gymnasium. The method was fully sampling-based, so I did not use the transition model

Method

Monte Carlo control learns from complete sampled episodes. After generating an episode with an $\epsilon - soft\ policy$, I computed the returns backward and updated Q(s,a)$Q(s,a)$ using the first visit of each state-action pair in that episode. The policy was then improved using an $\epsilon\epsilon$-greedy rule so the agent could keep exploring while gradually favoring better actions.

Results

For $\epsilon = 0.1\epsilon = 0.1$, the final greedy policy achieved a success rate of **0.185**, an average return of **0.185**, and an average successful episode length of **21.8** steps. For $\epsilon = 0.3\epsilon = 0.3$, the final greedy policy achieved a success rate of **0.515**, an average return of **0.515**, and an average successful episode length of **30.9** steps.



Comparison

The $\epsilon = 0.3\epsilon = 0.3$ setting performed much better than $\epsilon = 0.1\epsilon = 0.1$. This suggests that stronger exploration was important in the slippery FrozenLake environment, because the agent needed more exploration to discover better state-action paths.

Conclusion

This assignment showed that first-visit Monte Carlo control can learn an effective policy directly from sampled experience without access to the model. In this experiment, higher exploration led to better final performance.

Figures

- Figure 1: Success Rate vs Training Episodes
- Figure 2: Average Return vs Training Episodes

Disclaimer

Code debugging and report writing assisted with generative AI